

Taeyun Roh

<https://txxnrd.github.io>

Education

- **Korea University**, Seoul, Republic of Korea *Feb. 2026 – Present*
 - Master of Science in Computer Science and Engineering
- **Korea University**, Seoul, Republic of Korea *Mar. 2022 – Feb. 2026*
 - Bachelor of Science in Computer Science and Engineering

Research Experience

- **Data Mining & Information Systems Laboratory, Korea University**
Advisor: Professor Jae Woo Kang *Mar. 2025 – Present*
 - Research on agentic memory systems for AI agents
- **Network and Mobile System Laboratory, KAIST**
Advisor: Professor Sung-Ju Lee *Jan. 2025 – Mar. 2025*
 - Focused on efficient LLM inference, including on-device LLM inference
 - Developed LLM-based methods to detect the malicious Spymic app, enhancing threat identification accuracy
- **System Software Laboratory, Korea University**
Advisor: Professor Gyeongsik Yang *Jan. 2024 – Dec. 2024*
 - Research on federated learning in the healthcare domain with awareness of resource heterogeneity

Publications

- **CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents**
 - Under Review

Development & Related Experience

- **Aigensciences** *AI Researcher, Mar. 2025 – Aug. 2025*
 - Constructed an LLM-based pipeline for automated hypothesis generation in drug discovery
 - Developed a specialized pediatric counseling chatbot designed for KakaoTalk integration
- **Blurting** *Frontend Developer, Jun. 2023 – Jan. 2025*

- Co-founded a 4-member startup within a student club and took full responsibility for the entire frontend development
- Developed and operated a blind dating application with a novel matching format, achieving a peak ranking of 73rd in the App Store Social Networking category in Korea
- **Hiconsy** *Data Analyst, Jan. 2022 – Jan. 2023*
 - Developed mathematical models to optimize the Teaching Assistant (TA) compensation system and used regression analysis to investigate the correlation between student study time and academic performance

Awards

- **1st Place, Korea University Capstone Design** *Prize: ₩1,800,000*
 - Developed *NEWSQRAB*, an AI-driven system that automates news-to-short-form video generation using RAG-based LLMs
- **2nd Place, Korea University NEXT Software Competition** *Prize: ₩1,000,000*
 - Recognized for technical excellence as the Lead AI Engineer for the *NEWSQRAB* project
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** *Dec. 2024*
 - Awarded for the paper: “*Analysis on Research for Data and Resource Heterogeneity in Split-Federated Learning*”
- **2nd Place, 2023 Korea University Hackathon** *Prize: ₩1,000,000*
 - Developed *Noonchi*, a real-time visitor prediction and recommendation service for Han River parks based on user preferences

Projects

- **Xalute: Wearable Health Monitoring App with Korea University Anam Hospital**
 - Collaborated with Korea University Anam Hospital to develop a digital healthcare mobile application that interfaces with wearable devices, enabling real-time health tracking and intelligent alerts
- **Kakao Pediatric Response Chatbot**
 - Co-developed a pediatric Q&A chatbot system in collaboration with Kakao Healthcare and Upstage, enabling parents to receive real-time responses for child healthcare concerns using large language models

Languages

- **English:** TOEIC 940